

# Finwiser — Technical Documentation

HyKr Build Challenge 2026 · MVP Phase Submission Last updated: 2026-05-13 · Author: Chandrachuda Sarma, Founder A detailed engineering appendix with full module deep-dives, formula derivations, and operational runbook ( `Finwiser_Technical_Overview.md` ) is browsable alongside the production source at the **Codebase Explorer** → [finwiser.org/codebase/](https://finwiser.org/codebase/).

SEBI Registered Investment Adviser — INA000021331 · FIU-registered entity · Sahamati ReBIT 2.1.0 certified AA integration · OCI Mumbai + MongoDB Atlas Mumbai · Flutter Android in Google Play Closed Testing

## 1. Product Summary

Finwiser is a working Android reference app on a production backend that:

- Ingests consented multi-source financial data — Account Aggregator, Gmail, PDF statements, credit bureau.
- Normalises it into a unified ledger.
- Runs categorisation, cashflow analytics, and per-scheme mutual-fund intelligence.
- Produces goal-linked allocation recommendations under SEBI RIA suitability.

All financial logic runs server-side; the Flutter Android app is a thin reference client.

## 2. Tech Stack

| Layer          | Stack                                                                                                                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobile         | Flutter ≥ 3 / Dart ≥ 3 · <code>provider</code> (state) · <code>get_it</code> (DI) · <code>dio</code> (network) · <code>hive</code> (offline) · <code>flutter_secure_storage</code> · <code>eventsourcing</code> (SSE) · <code>firebase_messaging</code> · <code>fl_chart</code> · Rive |
| Backend        | Node.js 20 LTS · Express 4 · Mongoose 8 · Bull (queues) · Redis (managed)                                                                                                                                                                                                              |
| Database       | MongoDB Atlas (Mumbai · M10+) · CSFLE for PAN / mobile / financial PII                                                                                                                                                                                                                 |
| Data sources   | Account Aggregator (Saafe + Finvu) · Gmail Push via Pub/Sub · PDF ( <code>pdf-parse</code> ) · Experian via IDSPay                                                                                                                                                                     |
| Market data    | AMFI instrument master · <code>mfapi.in</code> NAV history                                                                                                                                                                                                                             |
| Infrastructure | OCI Mumbai (Ubuntu 22.04 LTS) · PM2 supervisor                                                                                                                                                                                                                                         |
| Security       | TLS 1.2+ · mTLS via SahamatiNet · JWE/JWS ( <code>node-jose</code> ) · JWT + JWKS · Helmet · <code>express-rate-limit</code> (Redis)                                                                                                                                                   |
| Observability  | Pino JSON · Sentry · Prometheus · Datadog · Slack + PagerDuty alerts                                                                                                                                                                                                                   |

**Non-obvious picks. MongoDB over Postgres** — evolving FI shapes absorbed without migrations; CSFLE for PII. **OCI Mumbai over AWS Mumbai** — SEBI / RBI residency; predictable egress; free-tier MAU. **PM2 fork mode over containers** — single service, single region; no container overhead at current scale.

### 3. Architecture

Four logical modules within a single Node.js service, with explicit data-flow boundaries. The monolith is a deliberate choice at current scale — DI-enforced module boundaries keep responsibilities sharp without distributed-system overhead.

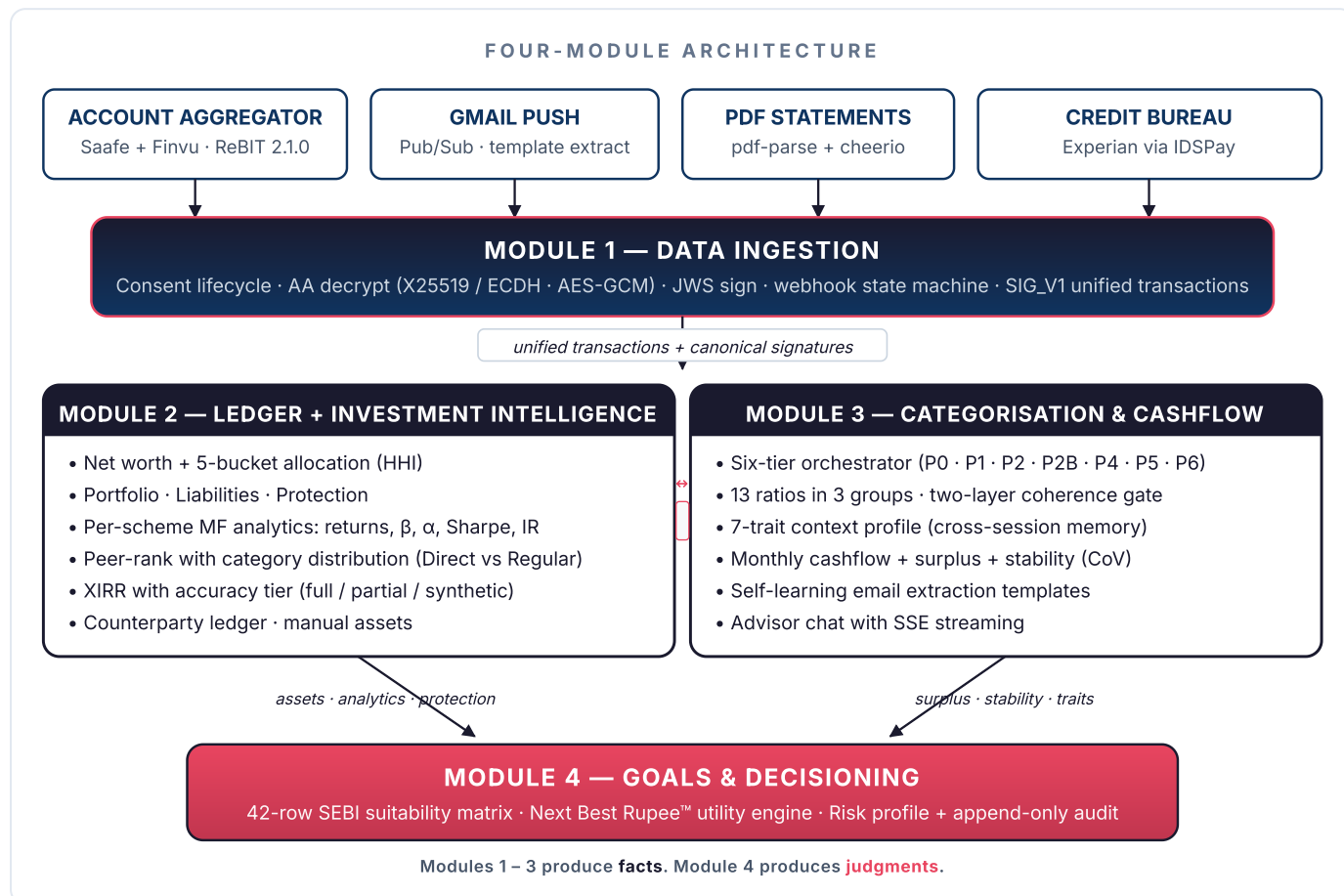


Figure 1 — Four-module architecture. Ingestion → Ledger × Categorisation → Decisioning.

- **Module 1 — Data Ingestion.** External acquisition: AA fetch lifecycle, Gmail Push, PDF parsing, credit-bureau pulls. Outputs a unified transaction stream with canonical signatures for cross-source dedup.
- **Module 2 — Ledger + Investment Intelligence.** Net worth, 5-bucket allocation, per-scheme MF analytics (returns,  $\beta$ ,  $\alpha$ , Sharpe, IR), peer rank, XIRR with accuracy tiering.
- **Module 3 — Categorisation & Cashflow.** Six-tier categorisation orchestrator (P0–P6), ratio engine with coherence gate, 7-trait context profile, advisor chat over SSE.
- **Module 4 — Goals & Decisioning.** Goal lifecycle, 42-row SEBI mutual-fund suitability matrix, Next Best Rupee allocation engine — ranks active goals by a structured multiplicative utility and emits named rationale codes ( `must_have_bias` , `emergency_underfunded` , `high_urgency` , `within_policy_bands` , `stable_surplus` , `progress_momentum` ) so every recommendation is auditable and explainable.

## Deployment view.

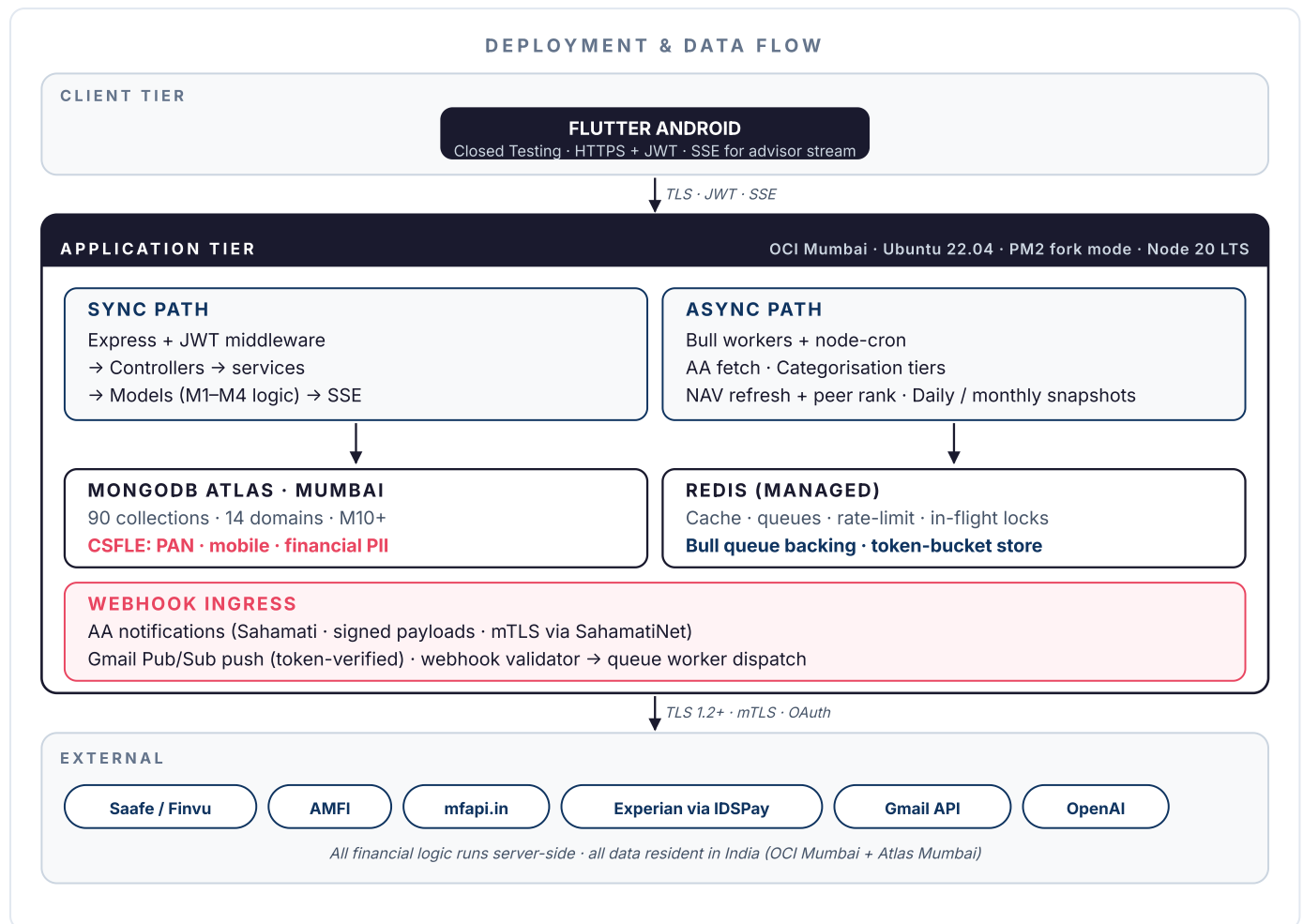


Figure 2 — Deployment topology and request paths (sync vs async, webhook ingress, external dependencies).

**Data model.** ~90 collections across 14 domains (AA pipeline, ledger, market data, categorisation, counterparty, goals, audit, etc.). Tiered TTL — 24h / 30d / 90d / ~7 years — depending on regulatory class.

## 4. Security & Compliance

Stack-level controls (TLS, mTLS, JWE/JWS, JWT/JWKS, Helmet, rate-limit, CSFLE) are listed in §2. The operational posture on top of that:

- **AA payload contract** — JWE (ECDH, AES-GCM) + JWS per ReBIT 2.1.0. Validated by the Sahamati certification harness (72/72 scenarios), which runs in CI on every backend PR.
- **CSFLE that cannot silently no-op.** The encryption plugin refuses to start in production without a configured master key, so "plaintext PII at rest" is structurally impossible — not a config-default risk.
- **Idempotency at every write boundary.** Cross-source transaction signatures, atomic `findOneAndUpdate` on snapshots, MongoDB-backed distributed locks for cron and webhook fan-out. Every state transition is replayable.
- **Two-tier audit trail.** Operational `calculationauditlogs` (90-day TTL) for decision replay; regulatory `legalretentionartifacts` (~7-year, pre-save immutable) for SEBI RIA Reg. 19.
- **Defence in depth.** Per-call PII audit logging; mock-credential refuse-to-boot guard in production; AA fair-use throttle enforced before the network call (not at the response).

## 5. Worked Example

One first connection, traced end to end. At each stop: the mechanisms that run — and what a naïve implementation gets wrong without them. A wrong number here isn't a bug; it's bad advice someone acts on with real money.

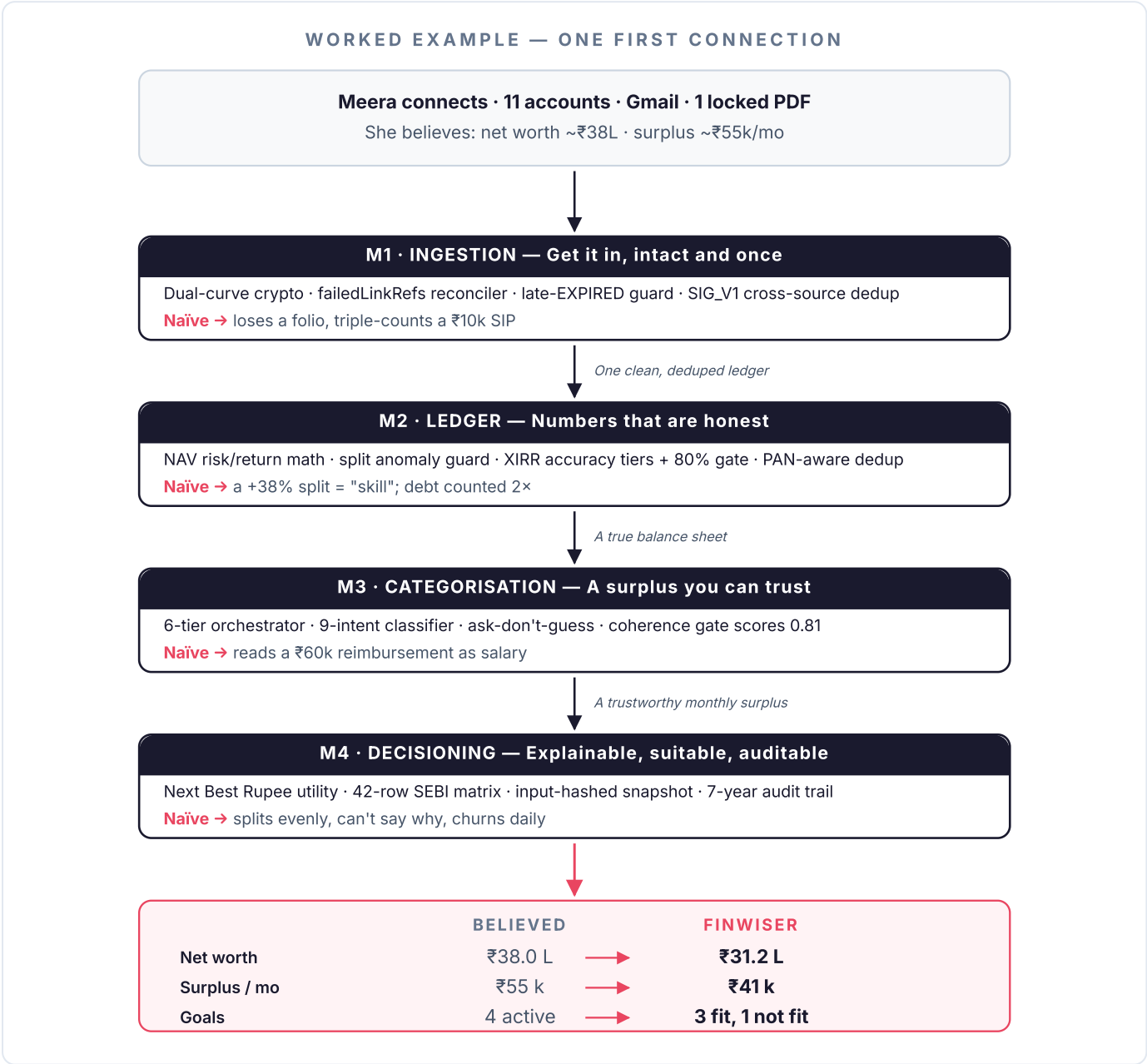


Figure 3 — One first connection, traced through all four modules: the mechanism at each stop, and what a naïve build gets wrong without it.

Meera asked one question — "where do I actually stand?" — and every step between her tap and the answer had a dozen ways to be quietly, confidently wrong. Closing that gap is what the engineering is for.

## 6. Deployment & Reviewer Access

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**Delivery.** SSH + `git pull` + `pm2 restart`. Health endpoint `GET /health`. Boot 3.5–8.5 s; rollback is `git revert` + redeploy.

**Environments.** Three tiers — local dev (Cloudflared exposes the dev host to AA sandbox), UAT / sandbox (isolated DB, sandbox vendors), and production. Database isolation enforced by `DB_NAME` (`main` / `finvu` / `saafe`).

**Scaling posture.** Single-node by design at current MAU. Distributed locks are MongoDB-backed, so adding a second OCI instance sharing Bull/Redis is an operational change, not an architectural one.

**Reviewer access.**

- **Codebase Explorer:** [finwiser.org/codebase/](https://finwiser.org/codebase/) — browse the full production backend (modules, services, models, routes) and the engineering appendix without a GitHub clone. Provided in lieu of a public repo link.
- **Android app:** Google Play Closed Testing → link in the submission form.
- **Test login:** mobile `9876543210` · OTP `123456`. The seeded test account bypasses the production onboarding flow so reviewers land directly on a populated ledger.
- **Live ingestion path.** Use the **Fresh AA fetch** action on the home screen to see the end-to-end pipeline (~60 s).

**Three steps after login:**

1. **Net Worth tab** → balance sheet, 5-bucket allocation, MF holdings with peer-percentile rank and XIRR accuracy tier.
2. **Goals tab** → tap any goal to see the Next Best Rupee rationale and structured factor breakdown.
3. **Advisor tab** → ask *"where am I spending the most?"* or *"should I increase my emergency fund?"* — responses stream via SSE.

**Technical questions:** `chandrachudasarma@gmail.com`

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## 7. Limitations

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- **Android only.** iOS deferred.
- **Closed Testing.** Not yet a public Play Store release.
- **Single region.** OCI Mumbai. DR documented; production is not multi-region.
- **Credit Bureau** (Experian via IDSPay) live since 2026-05-11 — early production volume.
- **AA throttle** of 45 calls per FIP per month applies under Sahamati fair-use during onboarding.

Full architecture, complete data model, cron schedule, retention tiers, security implementation detail, and decomposition history live in the **Detailed Architecture Appendix** (`Finwiser_Technical_Overview.md`), browsable via the **Codebase Explorer** → [finwiser.org/codebase/](https://finwiser.org/codebase/).

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